

FEBRUARY 13TH, 2025

NO, INPLACE_VECTOR SHOULDN'T HAVE AN ALLOCATOR

NEVIN LIBER
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PRESENTATION
P3636R0

PAPER
P3581R0



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LEWG LOVES TO RE-LITIGATE

LEWG LOVES TO RE-LITIGATE INPLACE_VECTOR

EXAMPLE: BAD_ALLOC, AGAIN

Wg21telecons2024/P0843#Library-Evolution-2024-01-30

- Agreed upon in Varna
- Almost re-litigated at Library Evolution Telecon 2024-01-30
 - No new technical information

- POLL: We want to revisit the status quo in the paper: “Inplace_vector throw a “bad_alloc” when exceeding max_size

SF	F	N	A	SA
3	4	2	5	4

- Outcome: No consensus for a change

TOKYO & P3160R0



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P3160R0 LEWG LITIGATION

Wg21tokyo2024/P3160

- We care about this case
- It should be a different vocabulary type, such as `basic_inplace_vector<T, N, A>`
- Doing this in `inplace_vector` makes it an overly complicated type for the vast majority of users.

- It is not worth the compilation costs, as the vast majority of users will not use this feature
 - Sidebar: P3062R0 C++ Should Be C++ supports this claim with statistics on the usage of PMRs

Query	# files	%
*	34.6M	100%
std::pmr	1.7k	0.0049%

P3160R0 LEWG LITIGATION

Polls

POLL: We should promise more committee time to pursuing “An Allocator-aware inplace_vector,” knowing that our time is scarce and this will leave less time for other work.

SF	F	N	A	SA
6	6	4	5	6

- Outcome: No consensus.
 - The room was not supportive of applying this paper to inplace_vector data structure.

TOKYO PLENARY

N4980 WG21 March 2024 Hybrid meeting Minutes of Meeting

Two objections to P0843R11

- One technical (constexpr)
 - Should be resolved by next plenary
 - (Most) P0843 authors agreed this needs to be resolved
- One on allocator support
 - LEWG chair informed WG21 it was already discussed
 - P0843 authors disagreed with waiting a meeting just to re-litigate this
 - On their own initiative, P3160 authors scheduled it for the next SG14 telecon



ST. LOUIS



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ST. LOUIS LEWG

- The results of the SG14 meeting were not presented.

ST. LOUIS PLENARY

N4985 WG21 June 2024 Hybrid meeting Minutes of Meeting

- This paper was delayed in Tokyo. Have all the concerns now been addressed?
 - Yes. The paper was changed to limit the types that are constexpr which makes it possible to implement.
 - And there is also implementation experience now.
- Objections in the room.

	In favour	Opposed	Abstain
In Person	54	1	13
Online	16	0	16
Total	70	1	29

- Motion passes.
 - Only one WG21 member opposed this design



**“UNLESS SOMETHING CHANGES WE SHOULD
NOT RE-LITIGATE PREVIOUS DISCUSSIONS”
- LEWG CHAIR**



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P3160R2 - WHAT REALLY CHANGED?



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P3160R2

Polls

POLL: If the embedded issues can be solved, would you be OK with adding the allocator template parameter to inplace vector?

SF	F	N	A	SA
6	8	1	3	0

- Being “OK with adding the allocator template parameter” is a far cry from being asked to re-litigate a strong plenary-approved WG21 design!
 - We already knew that some folks were “OK with adding the allocator template parameter”.
 - This is not new information!

P3160R2 SG14 RESULTS

- “A number of attendees voiced support for being able to have fine-grained memory control for allocator-aware objects stored within an `inplace_vector` whether or not they would use such a facility.”
 - How many?
 - If they aren’t using this C++11 facility how are they the intended audience?

- “Most of the attendees did not object to adding allocator support to `inplace_vector`”
 - That doesn’t mean it is their preferred design.
 - That doesn’t mean it is the right design.

NO, INPLACE_VECTOR SHOULDN'T HAVE AN ALLOCATOR

- We should not re-litigate this.
- There is no new information here.
- Make it a separate type, such as `basic_inplace_vector<T, N, A>`.
- Or maybe `clump_small_vector`, while different (models an indefinite number of elements as opposed to `inplace_vector`, which models up to N elements), meets this need, since it is already in the world of allocation.
 - Am willing to work with the authors of P3160 to put this forward for C++29.

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